

Lecture 3 - Logic Gates

CSC-220

Logic Circuits

- Application of Boolean Algebra
- Computers are built from electronic components called logic gates
- Logic gates perform Boolean operations
- Boolean logic connects:
 - True / False
 - On / Off
 - 1 / 0

Today

- AND gates
- OR gates
- NOT gates
- Combining gates
- Translating Boolean expressions into circuits
- Translating circuits into Boolean expressions
- Binary addition

Last Class

- Boolean values:
 - T = True
 - F = False
- Basic operators:
 - AND: $p \wedge q$
 - OR: $p \vee q$
 - NOT: $\neg p$
- Boolean expressions can be simplified using Boolean algebra
- Logically equivalent expressions:
 - Produce the same result
 - For every possible combination of inputs

Review

$$p \rightarrow q \equiv \neg p \vee q$$

p	q	$p \rightarrow q$
F	F	T
F	T	T
T	F	F
T	T	T

Quick Exercise

$$p \wedge (p \vee q)$$

Step 1: Look at the outer operation

- The outer operation is AND:

$p \wedge (\text{something})$

For the whole expression to be True, p must already be True.

- If p is False, then:

$\text{False} \wedge \text{anything} = \text{False}$

So right away, **if p is False, the entire expression is False.**

Step 2: What happens if p is True?

Now suppose:

$p = \text{True}$

The expression becomes:

$\text{True} \wedge (\text{True} \vee q)$

But:

$\text{True} \vee q = \text{True}$

regardless of what q is.

So we get:

$\text{True} \wedge \text{True} = \text{True}$

Therefore, whenever p is True, the whole expression is True.

Step 3: Compare the results

We have discovered:

If p is False $\rightarrow$ expression is False

If p is True $\rightarrow$ expression is True

So the entire expression always has the exact same value as p .

Truth Table

p	q	$p \vee q$	$p \wedge (p \vee q)$
F	F	F	F
F	T	T	F
T	F	T	T
T	T	T	T

Proof Examples:

<https://www.geeksforgeeks.org/engineering-mathematics/absorption-law-in-boolean-algebra/>

Logic Gates

- Hardware implementation of operations such as AND, OR, and NOT.

Logic	Computer
True	On
False	Off
True	1
False	0

- A logic gate is an electronic component
- It receives one or more inputs
- It produces an output
- The output depends on:
 - The input values
 - The type of gate
- Think of a gate as a very small decision-making device

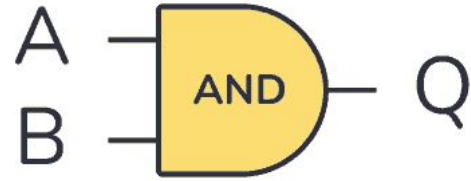
AND Gate

Inputs:

- $A = 1$
- $B = 1$

AND gate:

- Output = 1



A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

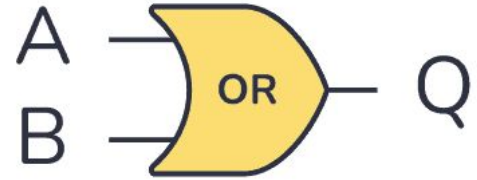
OR Gate

Inputs:

- $A = 0$
- $B = 1$

OR gate:

- Output = 1



A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

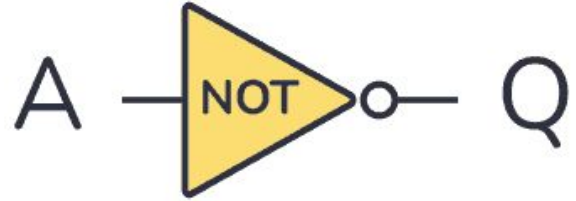
NOT Gate

Inputs:

- $A = 1$

NOT gate:

- Output = 0



A	Q
0	1
1	0

Other gates also exist

Examples:

- NAND
- NOR
- XOR

NAND:

$$\neg(A \wedge B)$$

NOR:

$$\neg(A \vee B)$$

XOR:

$$A \oplus B$$

Exclusive OR: XOR

$$A \oplus B \equiv (A \wedge \neg B) \vee (\neg A \wedge B)$$

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

Distributing NAND: $\neg(A \wedge B)$

- DeMorgan's Law
 - $\neg(p \wedge q) \equiv \neg p \vee \neg q$
 - $\neg(p \vee q) \equiv \neg p \wedge \neg q$
- $\neg(A \wedge B) \equiv \neg A \vee \neg B$

NAND Truth Table

A	B	$A \wedge B$	$\neg(A \wedge B)$	$\neg A \vee \neg B$
F	F	F	T	T
F	T	F	T	T
T	F	F	T	T
T	T	T	F	F

How to Read an Expression

Expression

$A \wedge (B \vee \neg C)$

Work from the inside outward

NOT C

- $\neg C$

OR with B

- $B \vee \neg C$

AND with A

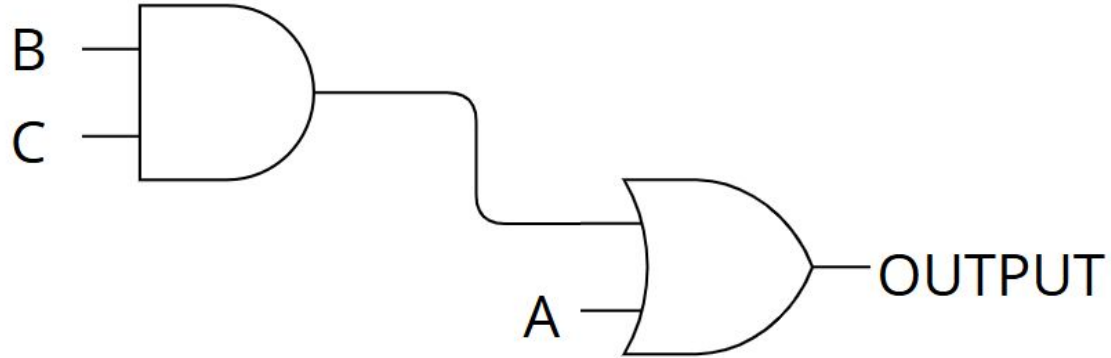
- $A \wedge (B \vee \neg C)$

Important

- Parentheses tell us how the gates should be connected

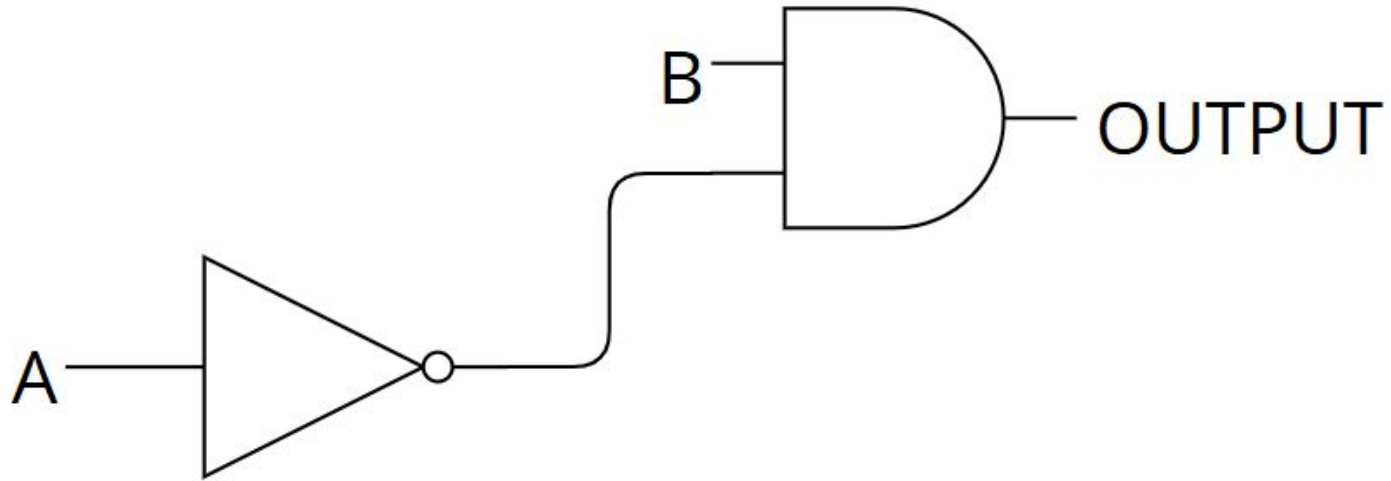
Exercise 1

Draw the Circuit for: $A \vee (B \wedge C)$

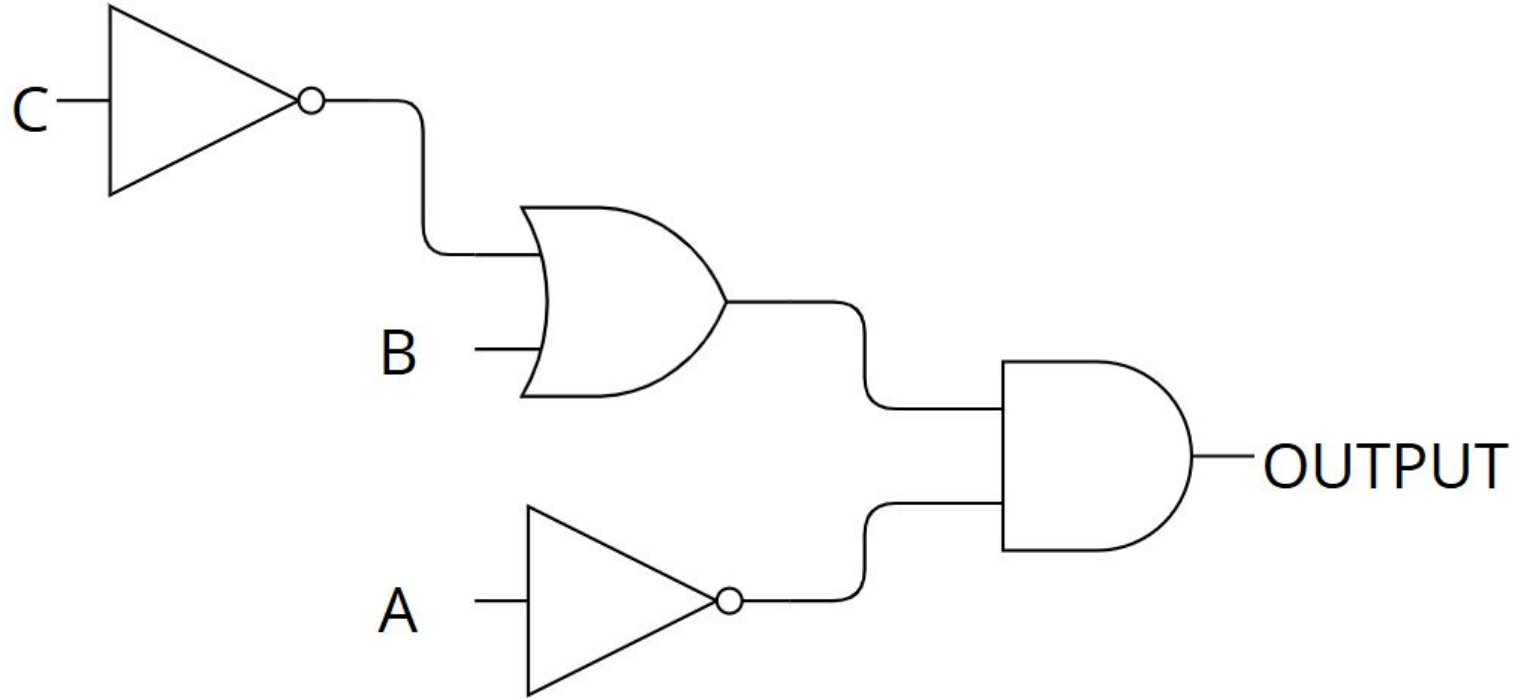


Exercise 2

Draw the circuit for: $\neg A \wedge B$



Circuit to Boolean Algebra



Answer

- $\neg A \wedge (B \vee \neg C)$
- $(\neg A \wedge B) \vee (\neg A \wedge \neg C)$

Why Simplify a Circuit?

Consider:

- $\neg A \wedge (B \vee \neg C)$
- $(\neg A \wedge B) \vee (\neg A \wedge \neg C)$

Which expression has fewer gates?

- $\neg A \wedge (B \vee \neg C)$

Fewer gates can mean:

- Simpler hardware
- Less physical space
- Less electrical power
- Fewer components
- Less complexity

Exercise 3: Simplify

$$(A \wedge B) \vee (A \wedge \neg B)$$

- Factor A

$$A \wedge (B \vee \neg B)$$

- $B \vee \neg B$
 - Law of Excluded Middle

B	$\neg B$	$B \vee \neg B$
F	T	T
T	F	T

- $B \vee \neg B \equiv T$
- $A \wedge T$
 - Identity Law

$$A \wedge T \equiv A$$

$$(A \wedge B) \vee (A \wedge \neg B) \equiv A$$

Original circuit:

$$(A \wedge B) \vee (A \wedge \neg B)$$

would require:

NOT gate

Two AND operations

OR operation

But the expression simplifies to:

A

Meaning:

The entire circuit is effectively just passing A through

B never actually changes the final result

Combinational Logic Circuits

- Has inputs
- Has outputs
- Contains no feedback loops
- Output depends on the current inputs
- Every combinatorial circuit can be represented using Boolean expressions

Basic Flow

Inputs → Logic gates → Output

Feedback Loop

A feedback loop occurs when:

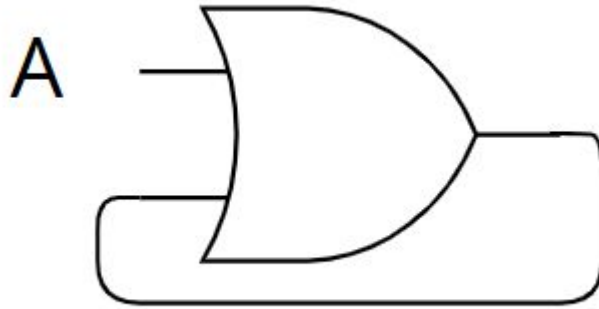
- The output eventually feeds back into an earlier input

These circuits behave differently from ordinary combinational circuits

Important idea:

- Feedback allows a circuit to maintain information
- This is important for computer memory

Example Circuit



A	Output
0	?
1	1

Disjunctive Normal Form

- Essentially ORs a group of ANDs
- $(A \wedge B) \vee (\neg A \wedge C)$

Truth Tables can Describe Circuits

Look at only rows with an output of 1

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

Row 2: $\neg A \wedge B$

Row 3: $A \wedge \neg B$

Combine using OR:

$(\neg A \wedge B) \vee (A \wedge \neg B)$

Exercise 4: Creating DNF

Create an expression for:

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	1

First True row:

$$\neg A \wedge \neg B$$

A and B must have the same value

Therefore it is a biconditional:

$$A \leftrightarrow B$$

Second True row:

$$A \wedge B$$

- "A if and only if B"

Combine:

$$(\neg A \wedge \neg B) \vee (A \wedge B)$$

Assignment/Homework

Pre-Class Survey

Read Chapter 1

Assignment 1 due 9/7 @ 11:59pm

Next Class

Binary Addition

Resources Used

<https://www.build-electronic-circuits.com/wp-content/uploads/2022/09/Truth-table-AND-gate.png>

<https://www.build-electronic-circuits.com/wp-content/uploads/2022/09/Truth-table-AND-gate.png>

<https://www.allaboutcircuits.com/textbook/digital/chpt-10/digital-logic-with-feedback/>